How-To Library

Informatica

Checklist to Automate Data Quality Rules in Axon Data Governance 6.2

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Abstract

In Axon Data Governance, you can automate the process of running data quality rules on Enterprise Data Catalog fields instead of manually generating the mappings and producing data quality scores. You can use Enterprise Data Catalog resources to scan for data objects, send the values of the data sources to Informatica Data Quality, and run rules on thousands of fields instantly. Axon retrieves the scores of the data quality rules from Informatica Data Quality and displays the scores for each field.

This document describes the process to automate data quality rules. It contains a checklist of the steps that you must perform so that Axon can onboard resources and fields from Enterprise Data Catalog and display data quality scores from Informatica Data Quality.

For details and guidelines on automating data quality rules, see the *Automate Data Quality Rules* chapter in the *Axon Data Governance User Guide*.

Supported Versions

- Informatica Axon Data Governance 6.2

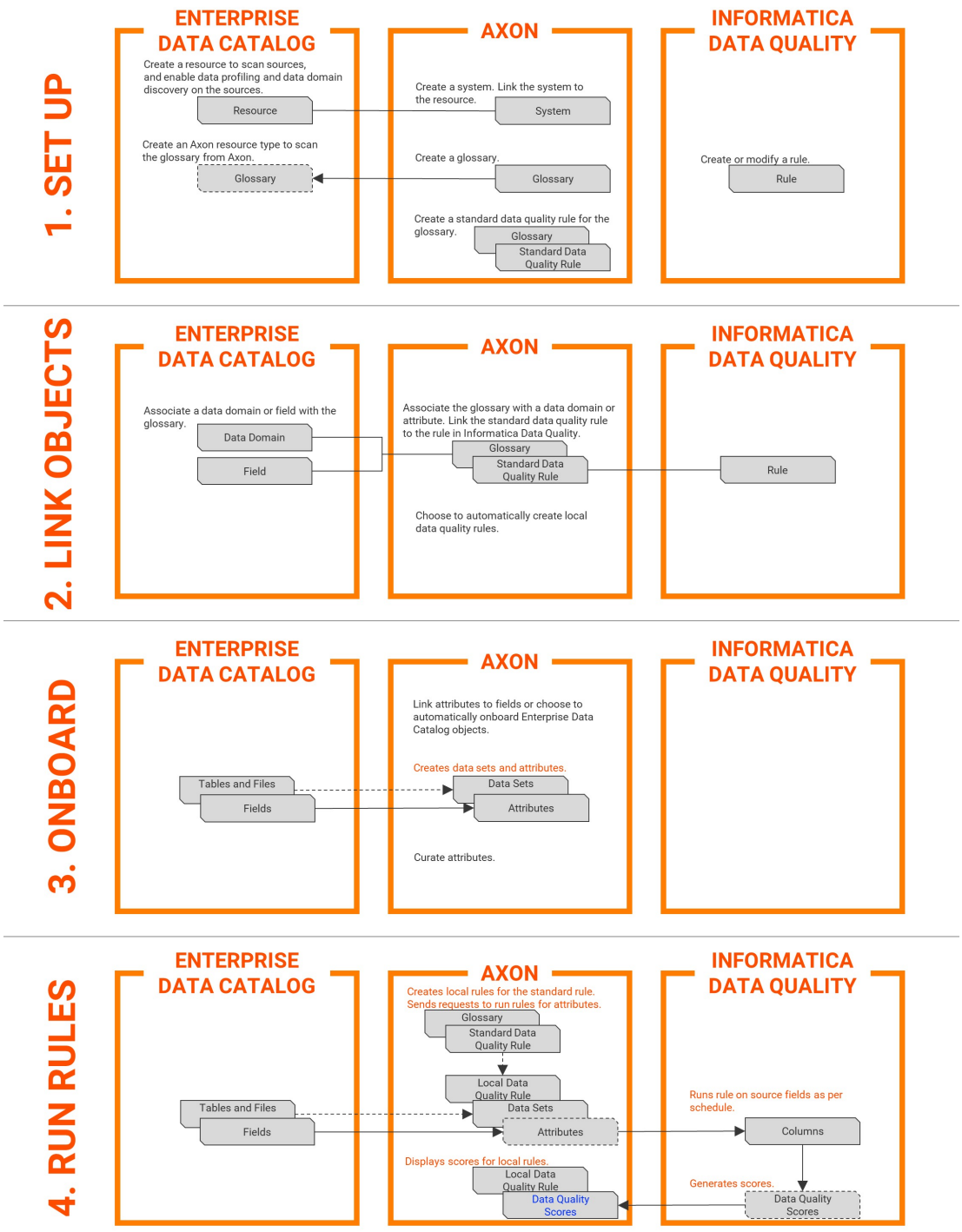
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Automating Data Quality Rules Process

The following image shows the steps required to set up Axon with Enterprise Data Catalog and Informatica Data Quality so that Axon runs the rules automatically on data store fields:



The following steps describe how to automate data quality rules:

1. Set up Axon, Enterprise Data Catalog, and Informatica Data Quality.
2. Link objects as required between Axon and Enterprise Data Catalog. Link objects as required between Axon and Informatica Data Quality.
3. Onboard objects from Enterprise Data Catalog to Axon. Optionally, manually create links between Enterprise Data Catalog fields and Axon attributes.
4. Run rules on the objects onboarded from Enterprise Data Catalog in Informatica Data Quality.

Step 1. Set Up Axon, Enterprise Data Catalog, and Informatica Data Quality

The first step to automate rules is to set up Axon, Enterprise Data Catalog, and Informatica Data Quality by creating the required objects.

Perform the following steps to create the objects:

1. In Enterprise Data Catalog, create a resource to scan data sources, and configure the resource to discover data profiles and data domains.
2. In Axon, create a System object that maps to the resource you created in Enterprise Data Catalog. Link the System object to at least one Enterprise Data Catalog resource to enable onboarding objects from Enterprise Data Catalog to Axon.
3. In Axon, create a Glossary object.
4. Create an Axon resource type in Enterprise Data Catalog. Use the Axon resource type to scan the Axon glossaries.
5. In Axon, create a standard data quality rule for the glossary objects. The standard data quality rule is a textual definition of the rule that you want to run on the fields.
6. In Informatica Data Quality, create rules to run on the fields. If the required rules already exist, modify them to conform to the rule automation guidelines

Step 2. Link Objects Between Axon, Enterprise Data Catalog, and Informatica Data Quality

The second step is to link the objects in Axon and Enterprise Data Catalog to onboard the fields, and link the objects in Axon and Informatica Data Quality to connect the rules.

Perform the following steps to link the objects:

1. In Axon, associate the Glossary object with the Enterprise Data Catalog data domains. In Enterprise Data Catalog, you can associate data domains and fields with the Axon glossary.
2. In Axon, link the standard data quality rule with a rule in Informatica Data Quality. This step associates a textual rule definition in Axon with a logical rule in Informatica Data Quality.
3. In Axon, choose the option to automatically create local data quality rules for the standard data quality rule. This step enables data quality rule automation.

Step 3. Onboard Objects from Enterprise Data Catalog to Axon

The third step is to display the objects from Enterprise Data Catalog in Axon. You can manually link the attributes in Axon with the fields in Enterprise Data Catalog. You can also choose to automatically onboard the objects from Enterprise Data Catalog to Axon. When you choose to onboard objects, the data sets and attributes are automatically created and onboarded to Axon.

Perform the following steps to onboard the objects:

1. In Axon, choose the option to automatically onboard the objects from Enterprise Data Catalog.
2. After the key elements are identified, the data sets and attributes are automatically created and onboarded to Axon. Axon is now ready to send the fields to Informatica Data Quality to run the rules.

3. In Axon, you can choose to accept or reject the Attribute objects created automatically.

Step 4. Run Informatica Data Quality Rules on Onboarded Objects

The fourth step is to run the rules automatically. Axon automatically runs Informatica Data Quality rules on Enterprise Data Catalog fields and displays the scores.

Axon performs the following steps to display the scores:

1. Axon creates a local data quality rule for the attributes that apply to each data set. In this step, Axon creates an instance of a rule for each field scanned by Enterprise Data Catalog. Axon then sends the rule mapping requests for the attributes to Informatica Data Quality.
2. Informatica Data Quality receives the mapping requests and runs the rule on the fields as per the schedule that you have defined.
3. Informatica Data Quality generates data quality scores for each field.
4. Axon retrieves the data quality scores for each table column and displays the scores for each local rule.

Checklist for Informatica Data Quality

Make sure that you set up Informatica Data Quality according to the following guidelines:

1. You have one of the following versions of Informatica Data Quality:
   - 10.2 HotFix 1
   - 10.2 HotFix 2
   - 10.2.2 HotFix 1
2. The following services are running on Informatica Data Quality:
   - Model Repository Service with monitoring enabled
   - Data Integration Service
   - Content Management Service
   - Scheduling service configured to the Model Repository Service
3. The Informatica Data Quality administrator has defined the rules according to the following specifications:
   - The output value is in the first column of the rule output.
   - The data type in the status output port of the rule is of string type.
   - The output value for the rule conforms to the value criteria that the administrator has configured in the Axon Agent.
4. Run the data quality rule on a test data set and verify that the output conforms to guidelines mentioned in the previous step.
5. Informatica Data Quality can connect to the sources scanned by Enterprise Data Catalog. Make sure that all critical connection parameters are configured in Informatica Data Quality, such as the password and Support Mixed-Case Identifier option for the resource.
6. Axon can connect to Informatica Data Quality. For more information, refer the *Configure Access to Informatica Data Quality* topic in the *Informatica Axon Data Governance 6.2 Administrator Guide*.
7. Axon connects to the Axon Agent using the HTTP protocol.
8. The database where Informatica Data Quality stores the data quality scores is a relational database.

Checklist for Enterprise Data Catalog

Verify the following configurations in Enterprise Data Catalog:

1. Enterprise Data Catalog runs on Informatica version 10.2.2 or higher.
2. Axon connects to Enterprise Data Catalog according to the following specifications:
   - You have entered all connection parameters correctly. For more information, refer the *Configure Access to Enterprise Data Catalog* topic in the *Informatica Axon Data Governance 6.2 Administrator Guide*.
   - In the **Admin Panel** > **Customize & Configure** > **System Settings** > **Group** > **Enterprise Data Catalog** > **Edit** page, select **Enable Automatic Onboarding**, and enter valid values for the **Confidence Score Threshold** and **Axon Super Admin Email** fields.
   - The source database that you want Enterprise Data Catalog to scan is Oracle, SQL Server, IBM DB2, or Teradata.
   - The file sources for which you want to run rules are CSV flat files in native file systems.
3. Link Enterprise Data Catalog resources to Axon systems. Perform the following steps to create the link:
   a. In Axon, go to **Create** > **Manage Catalog Links** > **Resources**.
   b. Right-click a resource, and select **Manage Linking** > **Create**.
   c. Enter the details of the system that you want to create, and click **Save**.

   The **Resources** facet displays the linked system for the resource in the **Linked Axon Systems** column. You can also open the System object and see the linked resource in the **Enterprise Catalog** tab.
4. Create data sets and attributes for the system that is linked to the Enterprise Data Catalog resource:
   a. In Axon, open the system object, and go to the **Enterprise Catalog** tab.
   b. Click **Edit** > **Physical Fields** to view the fields that Axon has retrieved from Enterprise Data Catalog.
   c. Right-click a field, and select **Manage Linking**.
   d. In the **Link Axon Attributes** dialog box, link to an existing attribute, or click **Create** to create an attribute.
   e. Select glossary terms that you want to associate with the attributes.
   f. Click **Save**.

Checklist for Axon and the Axon Agent

Axon

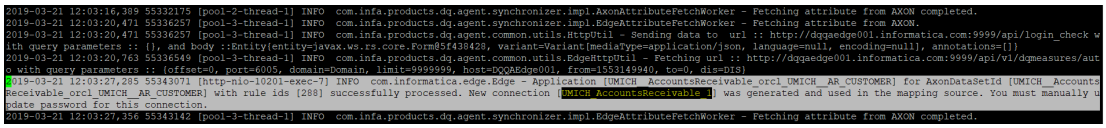
Configure Axon according to the following settings:

1. Enable rule automation. In the **Admin Panel** > **Customize & Configure** > **Application Settings** > **Data Quality** page, select the **Auto-generate local rules for standard rules** check box.
2. Create a standard data quality rule for the glossary object:
   a. Open a glossary object, click the **Edit** actions menu, and select **New Standard Data Quality Rule**.
   b. Enter the details of the standard rule, and set the **Axon Status** field to Active.
   c. In the **Technical Rule Reference** field, select a rule from Informatica Data Quality.
   d. Select the **Create Local Rules Automatically** check box.
   e. Click **Save** to associate the standard rule in Axon with the rule in Informatica Data Quality.

When you save the standard rule, Axon sends the metadata information of the Enterprise Data Catalog fields and Informatica Data Quality rule information to get the data quality scores. To view the scores, open the standard rule in Axon and go to the **Reports** tab. The **Latest Score** column displays the data quality scores for each local data quality rule.

Note: When Informatica Data Quality connects to an Enterprise Data Catalog resource for the first time, you must enter the password for the connection that you created in Informatica Data Quality. Open the Axon Agent log file to determine the connection that you created in Informatica Data Quality and the password that you must update. The Axon Agent file is `<Axon_Agent_Home>/logs/dq-axon/dqagent.log`.

The following image shows a sample Axon Agent log file:



Axon Agent

In the Axon Agent, specify the output values that must be used by Axon to automate data quality rules. If the output values do not conform to the value criteria, Axon displays the data quality scores as 0% and marks all row counts as exceptions.

For example, if Axon requires the output value to be either *Yes* or *No*, but the data quality rule generates the output value as *Y* or *N*, Axon displays the *Y* scores as exceptions.

To specify the output values for the rules, see the *Specify Output Values for Automated Rules* topic in the *Axon Data Governance 6.2 Administrator Guide*.

Author

Abhilash Scariya

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Pradeep G.N.